

Shubham S. Italiya

Final-year BSc Computer Science student | Python, SQL, AWS, TypeScript

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SUMMARY

Final-year Computer Science student at the University of East London with hands-on experience in data analysis, machine learning and full-stack web development. Skilled in Python (pandas, NumPy, scikit-learn, TensorFlow), SQL and AWS (Lambda, S3, Glue), and in TypeScript, JavaScript and Node.js from two years of remote full-stack development. Dissertation on time-series anomaly detection in financial transaction data using LSTM networks.

EDUCATION

BSc (Hons) Computer Science | University of East London, London | Jan 2024 - Sep 2027 (expected)

- Relevant modules: Machine Learning, Statistical Data Analysis, Databases & SQL, Cloud Computing, Software Engineering, Distributed Systems, Web Development, Data Structures & Algorithms
- Dissertation: Time-Series Anomaly Detection in Financial Transaction Data using LSTM networks
- Data Science Society: active member; took part in two internal machine learning hackathons

SKILLS

Programming languages: Python, SQL, TypeScript, JavaScript

Data science and machine learning: pandas, NumPy, scikit-learn, TensorFlow, Keras, Machine Learning, LSTM, ARIMA, Time-Series Forecasting, statsmodels, Anomaly Detection, Predictive Modelling, Feature Engineering, Statistical Analysis, SMOTE, imbalanced-learn

Data engineering and cloud: AWS Lambda, AWS S3, AWS Glue, AWS SNS, AWS SAM, CloudFormation, ETL, Data Pipelines, Data Quality Checks, Parquet, PostgreSQL, SQLite, Relational Databases, Data Cleaning

Web and software engineering: Node.js, FastAPI, SQLAlchemy, Jinja2, REST APIs, Git, GitHub, Code Reviews, Pair Programming, Testing, pytest, GitHub Actions, Debugging, Log Analysis, Production Support

Reporting and tools: Microsoft Excel, PowerPoint, Data Visualisation, Dashboards, Reporting, GitHub Copilot, ChatGPT, Generative AI tools

EXPERIENCE

Data Analyst (Part-Time) | RSVP Media Response Ltd, Canary Wharf, London | Dec 2025 - Feb 2026

- Collected, cleaned and reshaped raw data exports from accounting software with automated Python (pandas) pipelines, ensuring data accuracy and integrity before loading into management dashboards.
- Wrote and optimised SQL queries to extract key performance indicators (KPIs) from a multi-table relational database, reducing weekly report generation time by 40%.
- Prepared reports, dashboards and visual presentations of business performance metrics for management, translating data trends into plain-language, actionable recommendations.
- Built data visualisations and automated reporting in Excel and Python, reducing manual reporting effort and making insights easier to use for non-technical teams.

Full-Stack Developer (Remote) | Right Q Infotech, Remote | Oct 2023 - Dec 2025

- Developed and maintained production web services using TypeScript, JavaScript, Node.js and relational databases, delivering small, well-scoped changes that improved stability and user experience.
- Wrote maintainable, testable code supported by code reviews and pair programming, using Git/GitHub and GitHub Copilot to speed up delivery responsibly.
- Diagnosed and resolved live issues by investigating logs and SQL data, then documented fixes and follow-up actions to improve reliability and supportability.
- Worked with product and integration colleagues to understand requirements, test changes and support the onboarding of external API connections following documented processes.

PROJECTS

Cash Flow Forecasting Model (rebuilt 2026) | Python, TensorFlow, Keras, statsmodels, pandas, NumPy, AWS Lambda, AWS S3, github.com/shubham-italiya/cash-flow-forecasting | 2025; rebuilt Oct 2026

- Built a pipeline that forecasts the next 30 days of daily cash inflow for a UK online retailer, from about one million real transactions cleaned into a 739-day net sales series.

- Ran ADF stationarity tests on raw, log and differenced series and used them to set up a weekly seasonal ARIMA (SARIMA) model.
- Compared a seasonal-naive baseline, SARIMA and a TensorFlow LSTM in a 6-fold rolling backtest; the LSTM had the lowest daily MAE (9% below SARIMA, 2% below the baseline), while SARIMA was best on 30-day totals.

Customer Payment Behaviour Classifier (university group project, rebuilt 2026) | Python, scikit-learn, imbalanced-learn, pandas, SQL, SQLite, github.com/shubham-italiya/payment-behaviour-classifier | Oct 2024 - Dec 2024; rebuilt Oct 2026

- Built a model that flags customers likely to miss their next payment from six months of statement and payment history for 30,000 customers, reaching F1 0.54 and recall 0.57 on a held-out test set (34% higher F1 than a logistic-regression baseline).
- Engineered 15 behaviour features in SQL (joins and CASE aggregates) from normalised customer, statement and outcome tables.
- Compared SMOTE with class weights using stratified 5-fold cross-validation and tuned the decision threshold for F1; class weights won (PR-AUC 0.556 vs 0.528).

AWS Serverless Data Pipeline (rebuilt 2026) | Python, pandas, AWS Lambda, AWS S3, AWS SNS, AWS Glue, AWS SAM, CloudFormation, Parquet, moto, pytest, github.com/shubham-italiya/aws-serverless-data-pipeline | Mar 2025 - Apr 2025; rebuilt Oct 2026

- Built a serverless pipeline for daily transaction CSV files on AWS (S3, Lambda, SNS, Glue), with all infrastructure defined in one SAM/CloudFormation template that passes cfn-lint.
- Wrote 8 data-quality checks (schema drift, null rates, types, business rules, completeness, duplicates and volume anomalies with robust z-scores), with limits calibrated on 518 days of history; rejected batches are quarantined and trigger SNS alerts.
- Tested the pipeline end to end on a mocked AWS account (moto) with 52 real days and 7 deliberately broken files: all 7 broken files were caught and no real day was rejected.

Database Reporting Platform (rebuilt 2026) | Python, SQL, PostgreSQL, SQLite, SQLAlchemy, FastAPI, Jinja2, Chart.js, pandas, github.com/shubham-italiya/database-reporting-platform | Rebuilt Oct 2026

- Built a reporting platform on a 16-table relational database (Sakila sample, 16,044 rentals) with a FastAPI web dashboard: daily KPIs against a 7-day baseline, a revenue chart and a searchable, paged rentals table.
- Wrote the reporting queries as plain SQL (joins, aggregates and window functions such as RANK and SUM OVER) that run unchanged on SQLite and PostgreSQL.
- Automated a daily HTML e-mail report (Jinja2, matplotlib, SMTP) with top films, category revenue, a store comparison and an overdue-rentals list, saved alongside CSV files.